

Ai energy usage by query

Posted by u/asdfgayy • 0 points • 3 comments

probably this is a silly question, but are certain tasks 'harder' for llm ai (mostly I'm asking abt chatgpt) to do, not necessarily in the sense that it produces worse/less accurate results, just that it takes more energy.

If so, what kind of tasks are these? To my understanding AI just works on token prediction, so the perceived 'difficulty' of a task shouldn't matter for how much energy it uses. Between asking it to edit a text I wrote or generate its own text, to these tasks differ in energy cost? Is it based on how hard the text is to 'predict', the length of the response, or are all queries equal in this sense? thank you guys!

Comments

u/UnderstandingDry1256 • 1 points • 2026-02-17 14:28 UTC

Factor 1: size of the model. OpenAI likely uses some internal router which decides how smart the model should be to respond. Smarter model equals larger model equals more energy consumption.

Factor 2: thinking vs. non-thinking mode. Thinking models are actually agents which iterate over llm generating so-called like of thought, which also consumes tokens. The more iterations it needs to answer, the more energy it will consume in total.

u/[deleted] • 1 points • 2026-02-21 22:02 UTC

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u/Neinet3141 • 1 points • 2026-02-17 14:16 UTC

It depends. Each token is not equal.

One interesting example of why is KV caching, where it will re-use work done to generate previous tokens to generate the next token too, but this uses more memory.

Also, chatgpt does things like search the web which probably change the calculations, because it's being used in ways that aren't apparent to the user.

u/gift_for_aranaktu • 2 points • 2026-02-17 14:06 UTC

Yes, they can be - but a great deal of the energy footprint of AI is training the models, not necessarily just the cloud compute running them, so it's a bit more complicated.